

6.30. A sample of 10 television tubes produced by a company showed a mean lifetime of 1200 hours and a standard deviation of 100 hours. Estimate (a) the mean, (b) the standard deviation of the population of all television tubes produced by this company.

$$n=10$$

$$a) \mu = \bar{X} = 1200 \text{ hours}$$

$$b) s = 100 \quad \hat{s}^2 = \frac{n}{n-1} s^2$$

$$\hat{s}^2 = \frac{10}{9} (100)^2$$

$$\hat{s} = 105.4093 \text{ hours}$$

6.33. The mean and standard deviation of the diameters of a sample of 250 rivet heads manufactured by a company are 0.72642 inch and 0.00058 inch, respectively (see Problem 5.99). Find (a) 99%, (b) 98%, (c) 95%, (d) 90% confidence limits for the mean diameter of all the rivet heads manufactured by the company.

$$n=250 \quad \bar{X} = 0.72642 \quad s = 0.00058$$

$\therefore n > 30$, we can obtain critical value from table b-1

Confidence level	99%	98%	95%	90%
z_c	2.58	2.33	1.96	1.645

$$\text{Confidence intervals CI} = \bar{X} \pm z_c \frac{s}{\sqrt{n}}$$

$$a) 0.72642 \pm 2.58 \left(\frac{0.00058}{\sqrt{250}} \right) = 0.72642 \pm 0.000095 \text{ inch}$$

$$b) 0.72642 \pm 2.33 \left(\frac{0.00058}{\sqrt{250}} \right) = 0.72642 \pm 0.000085 \text{ inch}$$

$$c) 0.72642 \pm 1.96 \left(\frac{0.00058}{\sqrt{250}} \right) = 0.72642 \pm 0.000072 \text{ inch}$$

$$d) 0.72642 \pm 1.645 \left(\frac{0.00058}{\sqrt{250}} \right) = 0.72642 \pm 0.000060 \text{ inch}$$

6.35. If the standard deviation of the lifetimes of television tubes is estimated as 100 hours, how large a sample must we take in order to be (a) 95%, (b) 90%, (c) 99%, (d) 99.73% confident that the error in the estimated mean lifetime will not exceed 20 hours.

$$\mu = 100 \text{ hours} \quad \text{Error} \leq 20 \text{ hours}$$

$$a) 95\% \text{ confident} \quad z_c = 1.96$$

$$z_c \frac{\sigma}{\sqrt{n}} \leq 20 \quad \frac{1.96\sigma}{\sqrt{n}} \leq 20 \quad n \geq \left(\frac{1.96\sigma}{20} \right)^2 = 96.04$$

In order for us to be 95% confident that the error in the estimated mean

lifetime will not exceed 20 hours, n must 97 or larger.

b) 90% confident $z_c = 1.645$

$$\frac{1.645(100)}{\sqrt{n}} \leq 20 \quad n \geq \left(\frac{1.645(100)}{20} \right)^2 = 67.65$$

In order for us to be 90% confident that the error in the estimated mean lifetime will not exceed 20 hours, n must 68 or larger.

c) 99% confident $z_c = 2.58$

$$\frac{2.58(100)}{\sqrt{n}} \leq 20 \quad n \geq \left(\frac{2.58(100)}{20} \right)^2 = 166.41$$

In order for us to be 99% confident that the error in the estimated mean lifetime will not exceed 20 hours, n must 167 or larger.

b) 99.73% confident $z_c = 3.00$

$$\frac{3.00(100)}{\sqrt{n}} \leq 20 \quad n \geq \left(\frac{3.00(100)}{20} \right)^2 = 225$$

In order for us to be 99.73% confident that the error in the estimated mean lifetime will not exceed 20 hours, n must 225 or larger.

6.39. Five measurements of the reaction time of an individual to certain stimuli were recorded as 0.28, 0.30, 0.27, 0.33, 0.31 second. Find (a) 95%, (b) 99% confidence limits for the actual mean reaction time.

$n=5, < 30$ $\sigma=?$ t -distribution

$$\bar{X} = 0.298$$

$$s^2 = \frac{1}{5} \sum_{i=1}^5 (X_i - \bar{X})^2 = 0.000456$$

$$\hat{s}^2 = \frac{n}{n-1} s^2 = \frac{5}{4} s^2$$

$$\hat{s}^2 = 0.023875$$

$$a) 95\% \quad -t_{0.975} < \frac{(\bar{X} - \mu)\sqrt{n}}{\hat{s}} < t_{0.975}$$

$$CI = \bar{X} \pm t_{0.975} \frac{\hat{s}}{\sqrt{n}} < \mu < \bar{X} + t_{0.975} \frac{\hat{s}}{\sqrt{n}}$$

$$CL = \bar{X} \pm t_{0.975} \frac{\hat{s}}{\sqrt{n}} = 0.298 \pm 2.78 \left(\frac{0.023875}{\sqrt{5}} \right) = 0.298 \pm 0.0305$$

b) 99%

$$CL = \bar{X} \pm t_{0.995} \frac{\hat{s}}{\sqrt{n}} = 0.298 \pm 4.60 \left(\frac{0.023875}{\sqrt{5}} \right) = 0.298 \pm 0.0495$$

6.41. How large a sample of marbles should one take in Problem 6.40 in order to be (a) 95%, (b) 99%, (c) 99.73% confident that the true and sample proportions do not differ more than 5%?

$$n=60 \quad p=0.7 \quad \text{Error} \leq 0.05 \quad P \pm z_c \sqrt{\frac{pq}{n}}$$

a) 95% $z_c = 1.96$

$$z_c \sqrt{\frac{pq}{n}} \leq 0.05$$

$$z_c^2 \frac{pq}{n} \leq 0.05^2$$

$$n \geq \frac{z_c^2 pq}{0.05^2} = \frac{1.96^2 (0.7)(0.3)}{0.05^2} = 322.6944$$

n should be at least 323

b) 99% $z_c = 2.58$

$$n \geq \frac{2.58^2 (0.7)(0.3)}{0.05^2} = 559.1376$$

n should be at least 560

c) 99.73% $z_c = 3.00$

$$n \geq \frac{3^2 (0.7)(0.3)}{0.05^2} = 756 \quad n \text{ should be at least } 756$$

6.46. The standard deviation of the breaking strengths of 100 cables tested by a company was 1800 lb. Find (a) 95%, (b) 99%, (c) 99.73% confidence limits for the standard deviation of all cables produced by the company.

$$n=100 \quad s=1800 \quad CL = s \pm z_c \sigma_s \quad \because n=100 \quad \therefore \text{ sampling distribution is very}$$

near normal σ_s can be used $\sigma_s = \frac{\sigma}{\sqrt{2n}}$

$$\begin{aligned} \text{a) } 95\% \quad z_c &= 1.96 \quad CL = 1800 \pm 1.96 \left(\frac{1800}{\sqrt{2100}} \right) = 1800 \pm 249.4716 \\ \text{b) } 99\% \quad z_c &= 2.58 \quad CL = 1800 \pm 2.58 \left(\frac{1800}{\sqrt{2100}} \right) = 1800 \pm 328.3816 \\ \text{c) } 99.73\% \quad z_c &= 3.00 \quad CL = 1800 \pm 3.00 \left(\frac{1800}{\sqrt{2100}} \right) = 1800 \pm 387.8416 \end{aligned}$$

6.57. A population has a density function given by

$$f(x) = \begin{cases} (k+1)x^k & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

For n observations $X_1, \dots, X_n$ made from this population, find the maximum likelihood estimate of k .

$$\text{Likelihood function} \quad L = \prod_{i=1}^n f(x_i) = \prod_{i=1}^n (k+1)(x_i)^k \quad x_i \in [0, 1]$$

$$L = (k+1)^n \prod_{i=1}^n (x_i)^k$$

$$L = (k+1)^n (x_1 \cdot x_2 \cdots x_n)^k$$

$$\ln L = n \ln(k+1) + k \ln(x_1 \cdots x_n)$$

$$\frac{1}{L} \frac{dL}{dk} = \frac{n}{k+1} + \ln(x_1 \cdots x_n)$$

$$\text{let } \frac{dL}{dk} = 0 \quad k = -\frac{n}{\ln(x_1 \cdots x_n)} - 1$$

